

Integral Control of Oseen Flows With Mixed Boundary Conditions

Jetro Sihvonen, Nicolas Vanspranghe, Lassi Paunonen, Jan Heiland

Abstract—In this paper, we consider set-point output tracking for the Oseen linearisation of the incompressible Navier–Stokes equations around a stable equilibrium. We study the 2-D cylinder wake fluid model at low Reynolds numbers with boundary input and output, a model with mixed boundary conditions. We show that an integral controller tracks constant reference signals and rejects constant input disturbances. We illustrate the results using numerical simulations. These findings are a first step towards local output regulation of nonlinear flows.

Index Terms—Control of PDEs, Navier–Stokes equations, Oseen linearisation, Integral control, Mixed boundary conditions, Boundary control, Output tracking.

I. INTRODUCTION

The incompressible Navier–Stokes equations describe the behaviour of fluids in a large number of applications ranging from ship propellers to blood flows in the brain [1, 2]. Due to the complexity of fluid dynamics, the control of models governed by the Navier–Stokes equations and their linearisations is of great interest [3–5]. Significant attention has been dedicated to well-posedness, controllability and stabilisation of different kinds of models and setups in the past decades and also very recently [6–11].

Output regulation, where the goal is to track a measurement of a system to a given reference signal, has been studied relatively little for the Navier–Stokes equations and, more generally, nonlinear PDEs. Most notably, approximate regulation of incompressible Navier–Stokes flows around a stable equilibrium was studied in [12] for a setup with homogeneous Dirichlet boundaries, point observation and in-domain control. Regulation of other nonlinear PDEs has been considered in [13–16]. Meanwhile output regulation of linear PDEs has been investigated extensively [17–21].

We consider the problem of output tracking for a classical benchmark setup with mixed boundary conditions, the 2-D cylinder wake model. Stabilisation of the model and related simulations have been studied in detail in [22–26], while output regulation has not yet been addressed. We solve a set-point output tracking problem using integral control in the presence of a constant disturbance for the Oseen

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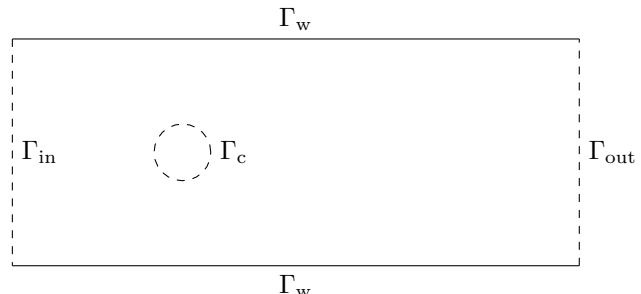


Fig. 1. Geometry of the domain.

linearisation around a stable equilibrium. Integral control is used extensively in industry due to its simplicity and robustness, and has been successfully applied in the context of nonlinear PDEs recently in [15, 16].

II. PROBLEM SETUP

The behaviour of the fluid is governed by the controlled incompressible Navier–Stokes equations

$$\frac{\partial v}{\partial t} - \nabla \cdot \sigma(v, p) + (v \cdot \nabla)v = 0, \text{ in } \Omega \times (0, \infty) \quad (1a)$$

$$\nabla \cdot v = 0, \text{ in } \Omega \times (0, \infty) \quad (1b)$$

$$v = n(x)g(x)u(t), \text{ on } \Gamma_c \times (0, \infty) \quad (1c)$$

$$v = f(x), \text{ on } \Gamma_{in} \times (0, \infty) \quad (1d)$$

$$v = 0, \text{ on } \Gamma_w \times (0, \infty) \quad (1e)$$

$$\sigma(v, p)n = 0, \text{ on } \Gamma_{out} \times (0, \infty) \quad (1f)$$

$$\Pi v(0) = v_0, \text{ in } \Omega. \quad (1g)$$

Here v is the velocity field, p is the pressure, $f : \Gamma_{in} \rightarrow \mathbb{R}^2$ is the profile of the inflow, $g : \Gamma_c \rightarrow \mathbb{R}$ is the profile of the control surface, n is the outward unit normal to the boundary and Π is the Leray projection, which projects onto the standard divergence free state space and v_0 is the initial velocity field. The Cauchy stress tensor σ is defined as

$$\sigma(v, p) := \nu(\nabla v + \nabla v^T) - pI,$$

where $\nu > 0$ is the viscosity. Hence we have $\nabla \cdot \sigma(v, p) = \nu \Delta v - \nabla p$. The system is controlled by $u(t) \in \mathbb{R}$, $t \geq 0$.

The domain $\Omega \subset \mathbb{R}^2$ is a rectangle with a disk cut out as shown in Fig. 1. This setup satisfies the assumptions in [22, H1-H4, H6], to which we refer the reader for many technical details. The condition (1f) is a Neumann type boundary condition, while the other boundary conditions are Dirichlet conditions, including the control boundary condition.

We will denote by $L^2(\mathcal{O})$ the Lebesgue spaces (of equivalence classes of) square integrable functions on a measurable

set $\mathcal{O} \subset \mathbb{R}^n$. Furthermore we denote by $H^s(\mathcal{O})$ the fractional Sobolev spaces of orders $s \in \mathbb{R}$; see [27] for precise definitions and norms.

We consider the boundary output

$$y(t) = Cv(\cdot, t) := \frac{1}{|\Gamma_{\text{out}}|} \int_{\Gamma_{\text{out}}} v(x, t) \cdot n(x) dx, \quad t \geq 0,$$

and a constant reference signal $y_{\text{ref}} \in \mathbb{R}$. The goal is to design a controller that achieves *exponential output tracking*, that is,

$$|e(t)| \leq Me^{-\alpha t},$$

for some $\alpha > 0$, $M > 0$, where $e(t) := y_{\text{ref}} - y(t)$, $t \geq 0$ is the *tracking error*. Before linearising the system we introduce a steady-state assumption.

Assumption 2.1: For $u(t) = 0$ there exists an equilibrium state $(v_*, p_*) \in H^{3/2+\varepsilon_0}(\Omega, \mathbb{R}^2) \times H^{1/2+\varepsilon_0}(\Omega)$ of (1) where the regularity index $0 < \varepsilon_0 < 1/2$ is as in [22, H5]. Away from the corners (v_*, p_*) has $H^2 \times H^1$ -regularity.

Remark 2.2: This assumption is in line with [22, H5] and is relevant in guaranteeing well-posedness. The weaker regularity compared to $H^2 \times H^1$ arises due to the corners. At low enough Reynolds number the assumption can be expected to hold when $f \in H^{3/2}(\Gamma_{\text{in}}, \mathbb{R}^2) \cap H_0^1(\Gamma_{\text{in}}, \mathbb{R}^2)$; see [22, 28, 29].

We also need an assumption on the control profile g to ensure that we can influence the system.

Assumption 2.3: We have $g \in H^{1/2}(\Gamma_c)$ and $\int_{\Gamma_c} g < 0$

Remark 2.4: This assumption holds for example for non-positive and nonzero g that is sufficiently smooth. Physically this corresponds to jets pushing fluid into the domain from the control surface Γ_c .

We can write $y_{\text{ref}} = y_* + \delta y$ where $y_* = Cv_*$ and $\delta y \in \mathbb{R}$. We shift the system by considering $v_s = v - v_*$ and $p_s = p - p_*$. In the shifted variables (v_s, p_s) the *Oseen linearisation* [4] around (v_*, p_*) is given by

$$\frac{\partial v_s}{\partial t} - \nabla \cdot \sigma(v_s, p_s) + (v_* \cdot \nabla)v_s + (v_s \cdot \nabla)v_* = 0$$

$$\nabla \cdot v_s = 0, \text{ in } \Omega \times (0, \infty) \quad (2a)$$

$$v_s = n(x)g(x)u(t), \text{ on } \Gamma_c \times (0, \infty) \quad (2b)$$

$$v_s = 0, \text{ on } \Gamma_{\text{in}} \cup \Gamma_w \times (0, \infty) \quad (2c)$$

$$\sigma(v_s, p_s)n = 0, \text{ on } \Gamma_{\text{out}} \times (0, \infty) \quad (2d)$$

$$\Pi v_s(0) = v_0 - \Pi v_*, \text{ in } \Omega. \quad (2e)$$

We define the output of the shifted system as $y_s(t) = Cv_s(\cdot, t)$, $t \geq 0$. The error is then

$$e(t) = y_{\text{ref}} - y(t) = \delta y + y_* - y_s(t) - y_* = \delta y - y_s(t), \quad (3)$$

for $t \geq 0$. In order to consider disturbance rejection we introduce a constant input disturbance $d \in \mathbb{R}$ to the linearised setup by replacing (2b) with

$$v_s = n(x)g(x)(u(t) + d), \quad \text{on } \Gamma_c \times (0, \infty). \quad (4)$$

In future work we hope to tackle general inflow disturbances, but to avoid technical difficulties we choose this

simpler case. We use an integral controller given by

$$\dot{z}(t) = e(t) = \delta y - y_s(t), \quad z(0) = z_0 \quad (5a)$$

$$u(t) = kz(t), \quad (5b)$$

for all $t \geq 0$, where $z(t) \in \mathbb{R}$, $t \geq 0$, is the controller state and $k > 0$ is a gain parameter. Our main result in Theorem 5.3 shows that when the equilibrium (v_*, p_*) is stable, with sufficiently small k we achieve output tracking of arbitrary constant references y_{ref} for the linearised system (2).

N.B. From this point on we denote the state of the shifted system as (v, p) , the output as y and the initial state as v_0 , dropping the subscripts.

III. ABSTRACT FORM

We denote by $\mathcal{L}(X, Y)$ the space of bounded operators from X to Y and by $R(\lambda, A) := (\lambda - A)^{-1}$ the resolvent of the operator $A : \mathcal{D}(A) \subset X \rightarrow X$. Furthermore we write $\Gamma_d = \Gamma_{\text{in}} \cup \Gamma_c \cup \Gamma_w$. We follow [22, 23] to define relevant function spaces. First we consider the pivot space

$$V_0 = \{v \in L^2(\Omega, \mathbb{R}^2) \mid \nabla \cdot v = 0, v \cdot n = 0 \text{ on } \Gamma_d\},$$

where $v \cdot n$ is the normal trace of v , which by [5, Thm. III.2.2] is well defined in $H^{-1/2}(\Gamma_d)$ owing to the divergence free condition. The space V_0 is a Hilbert space with the L^2 -inner product since it is a closed subspace of $L^2(\Omega, \mathbb{R}^2)$. The orthogonal projector from $L^2(\Omega, \mathbb{R}^2)$ onto V_0 is the *Leray projector* and is denoted by Π . Moreover, we define

$$V_1 = \{v \in H^1(\Omega, \mathbb{R}^2) \cap V_0 \mid v = 0 \text{ on } \Gamma_d\},$$

which likewise is a Hilbert space. Identifying the pivot space V_0 with its dual V_0' and letting $V_{-1} = V_1'$ we have the chain of dense and continuous embeddings $V_1 \hookrightarrow V_0 \hookrightarrow V_{-1}$. The *Oseen operator* $A : V_1 \rightarrow V_{-1}$ is defined via duality

$$\langle Av, w \rangle_{V_{-1}, V_1} = a(v, w) - b_*(v, w), \quad \forall v \in V_1, \forall w \in V_1,$$

where

$$a(v, w) := \int_{\Omega} \nu(\partial_1 v_2 + \partial_2 v_1)(\partial_1 w_2 + \partial_2 w_1) dx$$

$$b_*(v, w) := \int_{\Omega} ((v_* \cdot \nabla)v) \cdot w + ((v \cdot \nabla)v_*) \cdot w dx.$$

The integral defining the bilinear form $b_*(\cdot, \cdot)$ makes sense because of Assumption 2.1. The Oseen operator A has a realization as an unbounded operator in V_0 with domain $\mathcal{D}(A) := \{v \in V_1 \mid Av \in V_0\}$. The operator A and its adjoint A^* with their domains can be written in explicit form (see [22, Section 2.4]), which we do not give but rather denote $V_2 := \mathcal{D}(A)$ and $V_{-2} := (\mathcal{D}(A^*))'$. We have the following inclusions

$$V_2 \hookrightarrow V_1 \hookrightarrow V_0 \hookrightarrow V_{-1} \hookrightarrow V_{-2},$$

with dense and continuous embeddings [22]. The even-ordered spaces are part of a Sobolev tower [30, Section II.5a]. The space V_{-2} is also a Hilbert space. Finally we define $H := \{v \in L^2(\Omega, \mathbb{R}^2) \mid \nabla \cdot v = 0\}$, with the L^2 -norm, for which $V_0 \subset H$.

It is not possible to introduce boundary control action on the space V_0 since it incorporates a homogeneous boundary condition on the control surface [4, p. 29]. Thus we follow [4, 22] and write the system in abstract form in the larger space V_{-2} . The operator A admits an extension defined on V_0 and mapping onto V_{-2} given by

$$\langle Ay, v \rangle_{V_{-2}, \mathcal{D}(A^*)} = \langle y, A^*v \rangle, \quad \forall y \in V_0, \quad \forall v \in \mathcal{D}(A^*).$$

We do not distinguish notationally between the original and extended operators, since they retain similar properties owing to the Sobolev tower structure. We also define a Dirichlet lifting map Φ as follows: For $\eta \in H^{1/2}(\Gamma_c, \mathbb{R}^2)$, $\Phi\eta = \phi$ where ϕ is the solution of

$$\begin{aligned} \lambda_0 \phi - \nabla \cdot \sigma(\phi, \pi) + (v_* \cdot \nabla)\phi + (\phi \cdot \nabla)v_* &= 0 \\ \nabla \cdot \phi &= 0 \text{ in } \Omega, \quad \phi = \eta \text{ on } \Gamma_c \\ \phi &= 0 \text{ on } \Gamma_w, \quad \sigma(\phi, \pi)n = 0 \text{ on } \Gamma_{\text{out}}. \end{aligned}$$

The constant λ_0 is chosen so that the inequality $((\lambda_0 I - A)v, v)_{L^2} \geq \frac{\nu}{2} \|v\|_{V_1}^2$ is satisfied. Due to [22, Thm. 2.17] the operator Φ is well-defined and $\Phi \in \mathcal{L}(H^{1/2}(\Gamma_c, \mathbb{R}^2), H)$. As an immediate corollary, we have $\Phi ng \in \mathcal{L}(\mathbb{R}, H)$. This follows from the smoothness of Γ_c and from Assumption 2.1, which ensures that $g \in H^{1/2}(\Gamma_c)$, and hence $ng \in H^{1/2}(\Gamma_c, \mathbb{R}^2)$.

Following [22, Thm. 2.18] we split the system (2) as

$$\Pi \dot{v}(t) = A \Pi v(t) + (\lambda_0 I - A) \Pi \Phi ng(u(t) + d), \quad (6a)$$

$$\Pi v(0) = v_0, \quad (6b)$$

$$(I - \Pi)v(t) = (I - \Pi)\Phi ng(u(t) + d), \quad (6c)$$

for $t \geq 0$. The equation (6a) is an evolution equation on V_{-2} and $B := (\lambda_0 I - A) \Pi \Phi ng \in \mathcal{L}(\mathbb{R}, V_{-2})$, see [22, p. 3023].

IV. WELLPOSEDNESS OF THE CLOSED-LOOP SYSTEM

In this section we show the well-posedness of the closed-loop system consisting of the system (6) and the controller (5) by using properties of analytic semigroups. In order to employ frequency-domain arguments, we consider the natural complexification of our spaces and operators, without change of notation (see [4, p. 31]). We have

$$\begin{aligned} y(t) &= C \Pi v(t) + C(I - \Pi)v(t) \\ &= C \Pi v(t) + C(I - \Pi)\Phi ng(u(t) + d), \quad \forall t \geq 0 \end{aligned} \quad (7)$$

and $C \in \mathcal{L}(H, \mathbb{C})$ due to [5, Thm. III.2.2]. Thus the feedthrough operator $D := C(I - \Pi)\Phi ng$ satisfies $D \in \mathbb{C}$.

Now we can write the closed-loop system of (6) and (5) in terms of $v_e := (\Pi v, z) \in V_{-2} \times \mathbb{C}$ as

$$\dot{v}_e(t) = A_e v_e(t) + B_e w, \quad t > 0, \quad (8a)$$

$$e(t) = C_e v_e(t) + D_e w, \quad t \geq 0, \quad (8b)$$

$$v_e(0) = (v_0, z_0), \quad (8c)$$

where we have denoted $w = (d, \delta y)$,

$$\begin{aligned} A_e &= \begin{bmatrix} A & kB \\ -C & -kD \end{bmatrix}, \quad B_e = \begin{bmatrix} B & 0 \\ -D & I \end{bmatrix}, \\ C_e &= [-C \quad -kD], \quad D_e = [-D \quad I]. \end{aligned}$$

Remark 4.1: In system-theoretic terms the observation operator C is bounded with respect to the state space V_0 and unbounded with respect to the weaker space V_{-2} , while the control operator B is bounded with respect to the state space V_{-2} and unbounded with respect to the state space V_0 .

Theorem 4.2: Under Assumptions 2.1 and 2.3, the operator A_e generates an analytic semigroup on $V_{-2} \times \mathbb{C}$.

Proof: First note that since $B \in \mathcal{L}(\mathbb{C}, V_{-2})$ and $D \in \mathbb{C}$, [30, Thm. III.2.10] directly yields the result provided that

$$A_C = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}$$

generates an analytic semigroup on V_{-2} . Recall that a closed linear and densely defined operator $A_0 : \mathcal{D}(A_0) \subset X \rightarrow X$ on a Hilbert space X generates an analytic semigroup if and only if there exists $c > 0$ and $\mu \in \mathbb{R}$ such that

$$\|R(\lambda, A_0)\|_{\mathcal{L}(X)} \leq \frac{c}{|\lambda - \mu|},$$

for all $\lambda \in \mathbb{C}_{\mu+} := \{\lambda \in \mathbb{C} \mid \operatorname{Re} \lambda > \mu\}$ [4, Thm. 1.13]. From [22, Thm 2.8] we know that A generates an analytic semigroup (on V_0 and thus also on V_{-2}). Hence there exists $\mu \geq 1$ such that the resolvent $R(\lambda, A)$ exists for every $\lambda \in \mathbb{C}_{\mu+}$ and there exists $a > 0$ such that

$$\|R(\lambda, A)\|_{\mathcal{L}(V_{-2})} \leq \frac{a}{|\lambda - \mu|}, \quad \lambda \in \mathbb{C}_{\mu+}. \quad (9)$$

Furthermore for $\lambda \in \mathbb{C}_{\mu+}$ it is straightforward to check that the resolvent of A_C exists and has the form

$$R(\lambda, A_C) = \begin{bmatrix} R(\lambda, A) & 0 \\ \frac{1}{\lambda} C R(\lambda, A) & \frac{1}{\lambda} I \end{bmatrix}.$$

To estimate $R(\lambda, A_C)$ we still need to estimate $C R(\lambda, A)$. Using the resolvent identity, for all $\lambda \in \mathbb{C}_{\mu+}$

$$\begin{aligned} \|C R(\lambda, A)\|_{\mathcal{L}(V_{-2}, \mathbb{C})} &\leq \|C R(\mu, A)\|_{\mathcal{L}(V_{-2}, \mathbb{C})} \\ &\quad + \|C R(\mu, A)\|_{\mathcal{L}(V_{-2}, \mathbb{C})} \|(\mu - \lambda) R(\lambda, A)\|_{\mathcal{L}(V_{-2})}. \end{aligned} \quad (10)$$

Since $R(\mu, A) \in \mathcal{L}(V_{-2}, V_0)$ and $C \in \mathcal{L}(V_0, \mathbb{C})$ we have $\|C R(\mu, A)\|_{\mathcal{L}(V_{-2}, \mathbb{C})} < \infty$. Due to (9) and (10) for $\lambda \in \mathbb{C}_{\mu+}$

$$\|C R(\lambda, A)\|_{\mathcal{L}(V_{-2}, \mathbb{C})} < \|C R(\mu, A)\|_{\mathcal{L}(V_{-2}, \mathbb{C})} (1 + a). \quad (11)$$

Using (9) and (11) we get for all $v_e = (q, z) \in V_{-2} \times \mathbb{C}$ and $\lambda \in \mathbb{C}_{\mu+}$ that

$$\begin{aligned} \|R(\lambda, A_C) v_e\|_{V_{-2} \times \mathbb{C}}^2 &= \left\| \begin{bmatrix} R(\lambda, A)q \\ \frac{1}{\lambda} C R(\lambda, A)q + \frac{1}{\lambda} z \end{bmatrix} \right\|_{V_{-2} \times \mathbb{C}}^2 \\ &\leq \|R(\lambda, A)q\|_{V_{-2}}^2 + \left\| \frac{1}{\lambda} C R(\lambda, A)q \right\|^2 + \left\| \frac{1}{\lambda} z \right\|^2 \\ &\leq \frac{b}{|\lambda - \mu|^2} \|v_e\|_{V_{-2} \times \mathbb{C}}^2, \end{aligned}$$

for some $b > 0$. Thus A_C generates an analytic semigroup, which completes the proof since it is densely defined. ■

Theorem 4.3: Let Assumptions 2.1 and 2.3 hold. For any $(v_0, z_0) \in V_0 \times \mathbb{C}$ the closed-loop system consisting of (2) and (5) has a unique solution $(v, z) \in C(\mathbb{R}^+, L^2(\Omega, \mathbb{C}^2) \times \mathbb{C})$.

Proof: Let $v_e(0) = (v_0, z_0) \in V_0 \times \mathbb{C}$. Since the operator A_e generates a C_0 -semigroup $T_e(t)$ by Theorem 4.2, there exists a unique solution to (8a). This solution is

$$v_e(t) = T_e(t)v_e(0) + \int_0^t T_e(t-s)B_e w ds, \quad t \geq 0, \quad (12)$$

and satisfies $v_e := (q_1, z) \in C(\mathbb{R}^+, V_0 \times \mathbb{C})$ [4, p. 14]. Let $q_2 = (I - \Pi)\Phi ng(kz + d) \in C(\mathbb{R}^+, L^2(\Omega, \mathbb{R}^2))$. Then $q_1 + q_2$ is a solution of (6) and thus of (2) by [22, Thm. 2.18] for $u(t) = kz(t)$. By [22, Thm 2.15] this solution is unique. Thus $(q_1 + q_2, z) \in C(\mathbb{R}^+, L^2(\Omega, \mathbb{C}^2) \times \mathbb{C})$ is the unique solution of the closed-loop system. ■

Remark 4.4: The precise definition of a solution to (2), for which we refer the reader to [22, Def. 2.14], is given in the sense of transposition.

V. CLOSED LOOP STABILITY AND TRACKING

In this section we introduce our main result, Theorem 5.3. We show that the closed-loop system (8) is exponentially stable and that the transfer function $P(\lambda) = C(\lambda - A)^{-1}B + D$ (of (6), (7)) which describes the input-to-output behaviour, satisfies $P(0) > 0$. Finally we use these to prove Theorem 5.3.

To focus our attention on output tracking we introduce a stability assumption.

Assumption 5.1: The semigroup generated by the Oseen operator A is exponentially stable in V_0 .

Remark 5.2: Like Assumption 2.1, this assumption is expected to hold with sufficiently small Reynolds numbers as suggested by experimental results [31]. We hope to investigate tracking without this assumption in later works by considering stabilisation in conjunction with our analysis.

Theorem 5.3: Under Assumptions 2.1, 2.3 and 5.1 there exists $k_* > 0$ such that for all $0 < k < k_*$ the integral controller (5) solves the output tracking problem: For all initial states $v_e(0) = (v_0, z_0) \in V_0 \times \mathbb{R}$, input disturbances $d \in \mathbb{R}$ and reference signals $y_{\text{ref}} \in \mathbb{R}$ the output y of the system (2) converges exponentially to y_{ref} .

Remark 5.4: The result is true also for systems with other choices of $C \in \mathcal{L}(H, \mathbb{R})$ for which $P(0) > 0$.

Theorem 5.5: Let Assumptions 2.1, 2.3 and 5.1 hold and suppose that $P(0) \neq 0$. There exists $\varepsilon^* > 0$ such that for $0 < \varepsilon \leq \varepsilon^*$ the system (8a) is exponentially stable in the space $V_{-2} \times \mathbb{C}$ with the choice $k = \varepsilon(P(0))^{-1}$.

Proof: By the Gearhart–Greiner–Prüss Theorem [30, Thm. V.1.11] a C_0 -semigroup $T_0(t)$ with generator A_0 on a Hilbert space X is exponentially stable if and only if $\mathbb{C}_{0+}$ is in the resolvent set and $\sup_{\text{Re } \lambda > 0} \|R(\lambda, A_0)\|_{\mathcal{L}(X)} < \infty$.

Since the semigroup generated by A on V_0 is exponentially stable by Assumption 5.1, the semigroup generated by A on V_{-2} is exponentially stable by properties of Sobolev towers. Therefore there exists $M > 0$ such that $\|R(\lambda, A)\|_{\mathcal{L}(V_{-2})} \leq M$ when $\lambda \in \overline{\mathbb{C}_{0+}}$.

We denote $k_i = (P(0))^{-1}$ so that $k = \varepsilon k_i$ and use a similarity transformation

$$Q := \begin{bmatrix} I & 0 \\ \varepsilon H & \varepsilon I \end{bmatrix}, \quad Q^{-1} = \begin{bmatrix} I & 0 \\ -\varepsilon H & \frac{1}{\varepsilon} I \end{bmatrix},$$

where $H := CA^{-1} \in \mathcal{L}(V_{-2}, \mathbb{C})$ since $A^{-1} \in \mathcal{L}(V_{-2}, V_0)$ and $C \in \mathcal{L}(V_0, \mathbb{C})$. We denote $\tilde{A}_e := QA_eQ^{-1}$. We have $\mathcal{D}(\tilde{A}_e) = V_0 \times \mathbb{C}$ since $Q^{-1} \in \mathcal{L}(V_0 \times \mathbb{C})$ and $\mathcal{D}(A_e) = V_0 \times \mathbb{C}$. For $v_e = (q, z) \in V_0 \times \mathbb{C}$ we compute

$$\begin{aligned} \tilde{A}_e v_e &= Q \begin{bmatrix} (A - \varepsilon k_i B H)q + k_i B z \\ -\varepsilon C q + \varepsilon^2 k_i D H q - \varepsilon k_i D z \end{bmatrix} \\ &= \begin{bmatrix} (A - \varepsilon k_i B H)q + k_i B z \\ \varepsilon^2 H q - \varepsilon I z \end{bmatrix} = (A_1 + A_2)v_e. \end{aligned}$$

Here we denoted

$$A_1 := \begin{bmatrix} A & k_i B \\ 0 & -\varepsilon I \end{bmatrix}, \quad A_2 := \begin{bmatrix} -\varepsilon k_i B H & 0 \\ \varepsilon^2 H & 0 \end{bmatrix}.$$

We will show that $I - R(\lambda, A_1)A_2$ is boundedly invertible for sufficiently small $\varepsilon > 0$ and then use

$$\begin{aligned} R(\lambda, \tilde{A}_e) &= ((\lambda - A_1)(I - R(\lambda, A_1)A_2))^{-1} \\ &= (I - R(\lambda, A_1)A_2)^{-1}R(\lambda, A_1). \end{aligned} \quad (13)$$

First we compute that for $\lambda \in \overline{\mathbb{C}_{0+}}$

$$\begin{aligned} R(\lambda, A_1)A_2 &= \begin{bmatrix} R(\lambda, A) & \frac{1}{\lambda + \varepsilon} k_i R(\lambda, A)B \\ 0 & \frac{1}{\lambda + \varepsilon} I \end{bmatrix} A_2 \\ &= \begin{bmatrix} \left(\frac{\varepsilon^2}{\lambda + \varepsilon} - \varepsilon \right) k_i R(\lambda, A)B H & 0 \\ \frac{\varepsilon^2}{\lambda + \varepsilon} H & 0 \end{bmatrix}. \end{aligned}$$

To shorten notation in the following parts we will drop norm subscripts where possible without risk of confusion.

For all $\lambda \in \overline{\mathbb{C}_{0+}}$ we have $\frac{1}{|\lambda + \varepsilon|} \leq \frac{1}{\varepsilon}$. Therefore for $v_e = (q, z) \in V_{-2} \times \mathbb{C}$ and $\lambda \in \overline{\mathbb{C}_{0+}}$ we get

$$\begin{aligned} \|R(\lambda, A_1)A_2 v_e\| &= \left\| \begin{pmatrix} \left(\frac{\varepsilon^2}{\lambda + \varepsilon} - \varepsilon \right) k_i R(\lambda, A)B + \frac{\varepsilon^2}{\lambda + \varepsilon} \end{pmatrix} H q \right\| \\ &\leq \left(\left(\frac{\varepsilon^2}{\varepsilon} + \varepsilon \right) k_i M \|B\| + \frac{\varepsilon^2}{\varepsilon} \right) \|H\| \|q\| \\ &\leq \varepsilon(2k_i M \|B\| + 1) \|H\| \|v_e\|, \end{aligned}$$

since $H \in \mathcal{L}(V_{-2}, \mathbb{C})$, $B \in \mathcal{L}(\mathbb{C}, V_{-2})$. Thus for $\lambda \in \overline{\mathbb{C}_{0+}}$

$$\|R(\lambda, A_1)A_2\|_{\mathcal{L}(V_{-2} \times \mathbb{C})} \leq \varepsilon(2k_i M \|B\| + 1) \|H\|.$$

We can choose ε^* such that for all $0 < \varepsilon \leq \varepsilon^*$ we have $\|R(\lambda, A_1)A_2\| \leq 1/2 < 1$ for $\lambda \in \mathbb{C}^+$. Therefore for $\lambda \in \overline{\mathbb{C}_{0+}}$ the operator $I - R(\lambda, A_1)A_2$ in the space $V_{-2} \times \mathbb{C}$ has a bounded inverse given by a Neumann series so that

$$\begin{aligned} \|(I - R(\lambda, A_1)A_2)^{-1}\| &= \left\| \sum_{n=0}^{\infty} (R(\lambda, A_1)A_2)^n \right\| \\ &\leq \sum_{n=0}^{\infty} \|R(\lambda, A_1)A_2\|^n \leq \sum_{n=0}^{\infty} \frac{1}{2^n} = 2. \end{aligned} \quad (14)$$

Finally for $\lambda \in \overline{\mathbb{C}_{0+}}$ and $v_e = (q, z) \in V_{-2} \times \mathbb{C}$ we have

$$\begin{aligned} \|R(\lambda, A_1)v_e\| &= \left\| R(\lambda, A)q + \frac{1}{\lambda + \varepsilon} (k_i R(\lambda, A)Bz + z) \right\| \\ &\leq M \|q\| + \frac{1}{\varepsilon} (M \|k_i B\| + 1) \|z\| \\ &\leq (M + \frac{1}{\varepsilon} (M \|k_i B\| + 1)) \|v_e\|. \end{aligned} \quad (15)$$

Using (13) with (14) and (15) we have

$$\begin{aligned} \sup_{\operatorname{Re} \lambda \geq 0} \|R(\lambda, \tilde{A}_e)\| &= \sup_{\operatorname{Re} \lambda \geq 0} \|(I - R(\lambda, A_1)A_2)^{-1}R(\lambda, A_1)\| \\ &\leq \sup_{\operatorname{Re} \lambda \geq 0} \|(I - R(\lambda, A_1)A_2)^{-1}\| \|R(\lambda, A_1)\| \\ &\leq \sup_{\operatorname{Re} \lambda \geq 0} 2(M + \frac{1}{\varepsilon}(M\|k_i B\| + 1)) < \infty. \end{aligned}$$

Thus we may use the Gearhart–Greiner–Prüss Theorem to conclude that the semigroup generated by $\tilde{A}_e$ is exponentially stable whenever $0 < \varepsilon \leq \varepsilon^*$. By similarity the semigroup generated by A_e is also exponentially stable on $V_{-2} \times \mathbb{C}$ whenever $0 < \varepsilon \leq \varepsilon^*$. ■

Lemma 5.6: The transfer function of (2) at zero $P(0) = C(-A)^{-1}B + D$ is positive.

Proof: For fixed $u \in \mathbb{R}$ consider $\phi = -A^{-1}Bu + (I - \Pi)\Phi ngu$, for which $C\phi = P(0)u$. Clearly $-A^{-1}Bu \in V_0$ and since $\Phi ngu \in H$, also $(I - \Pi)\Phi ngu \in H$. Therefore $\nabla \cdot \phi = 0$ and using Gauss's identity (see [5, Ex. II.4.3], which can be applied via a density argument) we have

$$0 = \int_{\Omega} \nabla \cdot \phi = \int_{\Gamma_w} \phi \cdot n + \int_{\Gamma_{\text{out}}} \phi \cdot n + \int_{\Gamma_c} \phi \cdot n. \quad (16)$$

By Assumption 2.3 for $u > 0$ we have

$$\int_{\Gamma_c} \phi \cdot n = \int_{\Gamma_c} ug(x)n(x) \cdot n(x)dx = u \int_{\Gamma_c} g(x)dx < 0.$$

Thus since $\phi = 0$ on Γ_w , by (16) we have

$$y = C\phi = \frac{1}{|\Gamma_{\text{out}}|} \int_{\Gamma_{\text{out}}} \phi \cdot n = -\frac{1}{|\Gamma_{\text{out}}|} \int_{\Gamma_c} \phi \cdot n > 0.$$

Therefore $u > 0$ implies $y > 0$ (and similarly $u < 0$ implies $y < 0$), which shows that $P(0) > 0$. ■

Proof of Theorem 5.3: By Lemma 5.6 $P(0) > 0$ and thus by Theorem 5.5 we can choose k_* such that for $0 < k \leq k_*$ the closed-loop operator A_e generates an exponentially stable semigroup $T_e(t)$. We take $v_e(0) \in V_0 \times \mathbb{R}$, $d \in \mathbb{R}$ and $y_{\text{ref}} \in \mathbb{R}$. We follow techniques used in [17, Thm. 3.4] and construct the tracking error e explicitly. For this it is sufficient to consider (8) since its construction guarantees that its output is the tracking error corresponding to (3).

Let v_e be the solution of (8). The properties of semigroups and (12) imply

$$\begin{aligned} v_e(t) &= T_e(t)v_e(0) + \int_0^t -\frac{d}{ds}T_e(t-s)A_e^{-1}B_e w ds \\ &= T_e(t)v_e(0) + T_e(t)A_e^{-1}B_e w - A_e^{-1}B_e w. \end{aligned}$$

Thus by (8b)

$$e(t) = C_e T_e(t)(v_e(0) + A_e^{-1}B_e w) + (-C_e A_e^{-1}B_e + D_e)w.$$

We have $B_e \in \mathcal{L}(\mathbb{C}^2, V_{-2} \times \mathbb{C})$ and $A_e^{-1} \in \mathcal{L}(V_{-2} \times \mathbb{C}, V_0 \times \mathbb{C})$. Denote $x_e = (x, x_c) = A_e^{-1}B_e w \in V_0 \times \mathbb{C}$ and consider $A_e x_e = B_e w$, which in block form is

$$\begin{bmatrix} A & kB \\ -C & -kD \end{bmatrix} \begin{bmatrix} x \\ x_c \end{bmatrix} = \begin{bmatrix} B & 0 \\ -D & I \end{bmatrix} \begin{bmatrix} d \\ \delta y \end{bmatrix}.$$

Thus we have

$$[-C \quad -kD]x_e = [-D \quad I]w$$

and

$$(-C_e A_e^{-1}B_e + D_e)w = -[-C \quad -kD]x_e + [-D \quad I]w = 0.$$

This then means that the tracking error is

$$e(t) = C_e T_e(t)(v_e(0) + A_e^{-1}B_e w). \quad (17)$$

Since $T_e(t)$ is exponentially stable, $\|T_e(t)\|_{\mathcal{L}(V_0 \times \mathbb{C})} \leq M e^{-\alpha t}$ for all $t \geq 0$ where $M, \alpha > 0$. Hence by (17)

$$|e(t)| \leq \|C_e\|_{\mathcal{L}(V_0 \times \mathbb{C}, \mathbb{C})} M e^{-\alpha t} \|(v_e(0) + A_e^{-1}B_e w)\|_{V_0 \times \mathbb{C}}.$$

Therefore the output tracking problem is solved with exponential convergence. ■

VI. SIMULATIONS

We demonstrate our results by using the simulation setup introduced in [25], implementing the controller (5) using discretisations in time and space. The simulations are actually performed on a numerical discretisation of the *nonlinear* Navier–Stokes equations. Thus they not only illustrate the results for the linearised system, but also suggest that the controller may be applicable to the nonlinear system as well. The choice of g in the simulations is such that the control is comprised of two jets. The inflow profile f is a fixed parabolic function. We checked that the output of the system y converges exponentially to a given reference value for different choices of parameters that can be expected to satisfy the Assumptions 2.1, 2.3 and 5.1.

We illustrate the simulated setup in Fig. 2 where the controlled output, $y_c(t)$, uncontrolled output $y(t)$ and the reference y_{ref} are shown. The control action (and thus also the control disturbance) is initiated after allowing the system to stabilise. In Fig. 3 a snapshot of the flow at time $t = 0.6$ is shown. The flow is non-turbulent and the control actions are visible as jets from the control surface. We considered $y_{\text{ref}} = 0.87$, $d = 0.1$, and $k = 41$ with a data grid of size $N = 19512$, and a time discretisation with $N_t = 2048$ for $t \in [0, 0.6]$. Here the gain k was chosen experimentally so that it is sufficiently small.

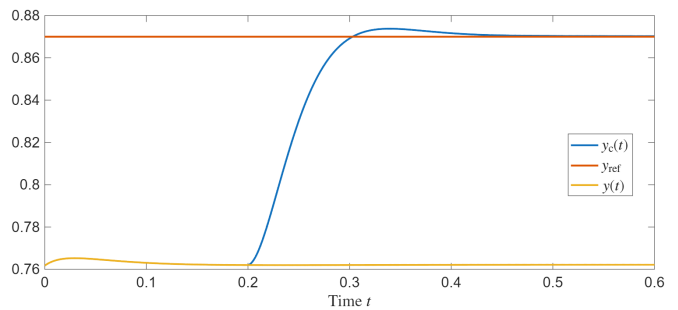


Fig. 2. Controlled output and reference

VII. CONCLUSIONS

We have shown that for the Oseen linearisation of the cylinder wake model (2) a simple integral controller (5) solves the output tracking problem under regularity and stability assumptions on the equilibrium. The controller tracks

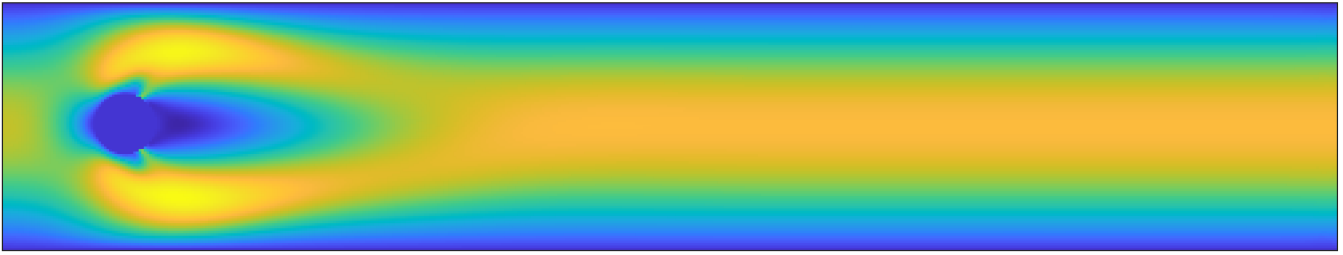


Fig. 3. Snapshot of the controlled cylinder wake model

constant reference signals exponentially and rejects constant input disturbances. Our results are a first step towards local output regulation of the Navier–Stokes equations for the cylinder wake model. We expect that the results for the linearisation may apply to the nonlinear model. Furthermore, stabilisation results in [9, 22] provide the possibility of constructing a PI-type controller which could achieve output regulation for higher Reynolds numbers, that is, when the equilibrium is not stable.

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